Distributed Systems

Abstractions for Time

Time - A Fundamental Concept (Recap)

"Time is what clocks measure. We use time to place events in sequence one after the other, and we use time to compare how long events last..." [Internet Encyclopedia of Philosophy].

In distributed systems, there are two concepts of time:

- Physical time measured by physical clocks.
  - Clocks tick with a certain rate.
  - Used to sequence events, determine durations of event, measure intervals between them, quantify rate of change such as motion of objects.
- Logical time measured by logical clocks.
  - Clocks tick when certain events occur, e.g., send or receive events.
  - Used to order causally related events and to detect causality violations.

When a system has *n* computers, all *n* crystals will run at slightly different rates…This difference in time values is called clock skew.

Physical Clocks (Recap)
Definitions

Let $C(t)$ be a perfect clock.

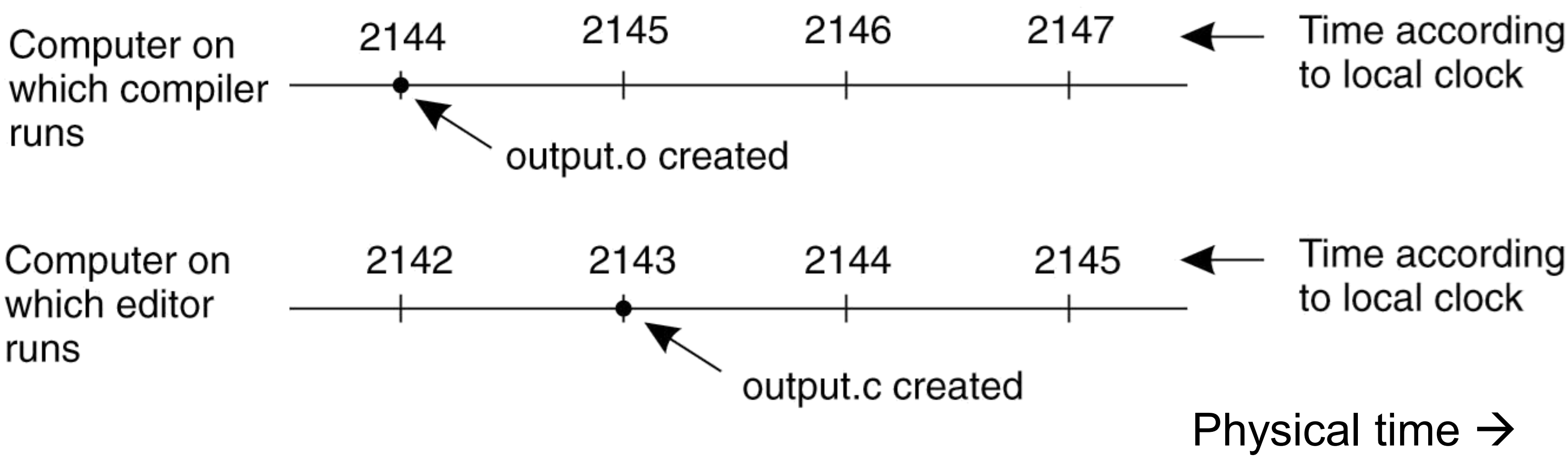
A clock $C_i(t)$ is called correct at time t if $C_i(t) = C(t)$.

A clock $C_i(t)$ is called accurate at time t if $dC_i(t)/dt = dC(t)/dt \equiv 1$.

Two clocks $C_i(t)$ and $C_k(t)$ are synchronised *at time t* if $C_i(t) = C_k(t)$.

These definitions only apply to physical clocks!

A distributed edit-compile workflow



2143 < 2144 ➔ make doesn’t call compiler

Lack of time synchronization result – a possible object file mismatch

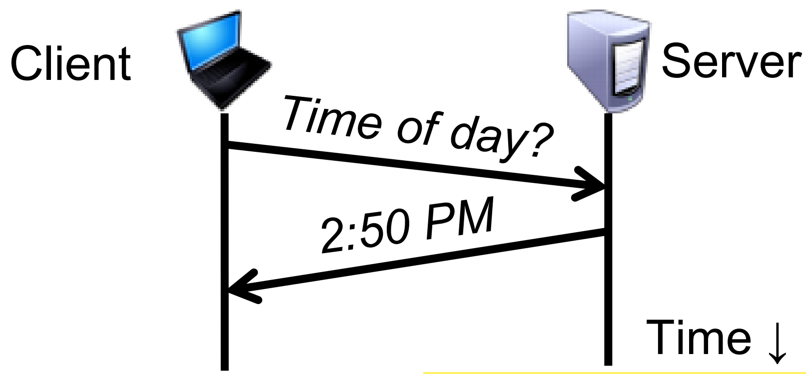
What makes time synchronization hard?

1. Quartz oscillator sensitive to temperature, age, vibration, radiation
   Accuracy ~one part per million
      (one second of clock drift over 12 days)

2. The internet is:
   Asynchronous: arbitrary message delays
   Best-effort: messages don't always arrive

Synchronization to a time server

- Suppose a server with an accurate clock (e.g., GPS-receiver)
  - Could simply issue an RPC to obtain the time:



- But this doesn't account for network latency
  - Message delays will have outdated server's answer

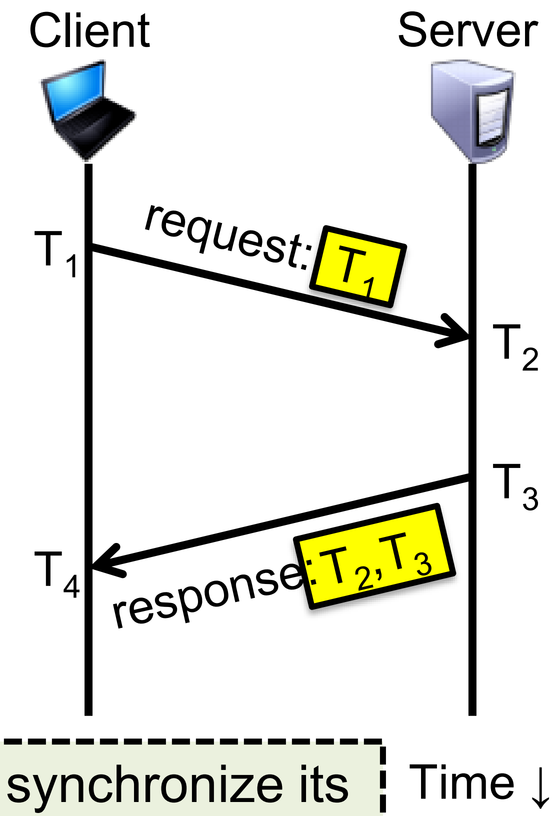
Cristian's algorithm: Outline

1. Client sends a request packet, timestamped with its local clock T_1

2. Server timestamps its receipt of the request T_2 with its local clock

3. Server sends a response packet with its local clock T_3 and T_2

4. Client locally timestamps its receipt of the server's response T_4



How can the client use these timestamps to synchronize its local clock to the server's local clock?

Cristian's algorithm: Offset sample calculation

Goal: Client sets clock ← $T_3 + \delta_{resp}$

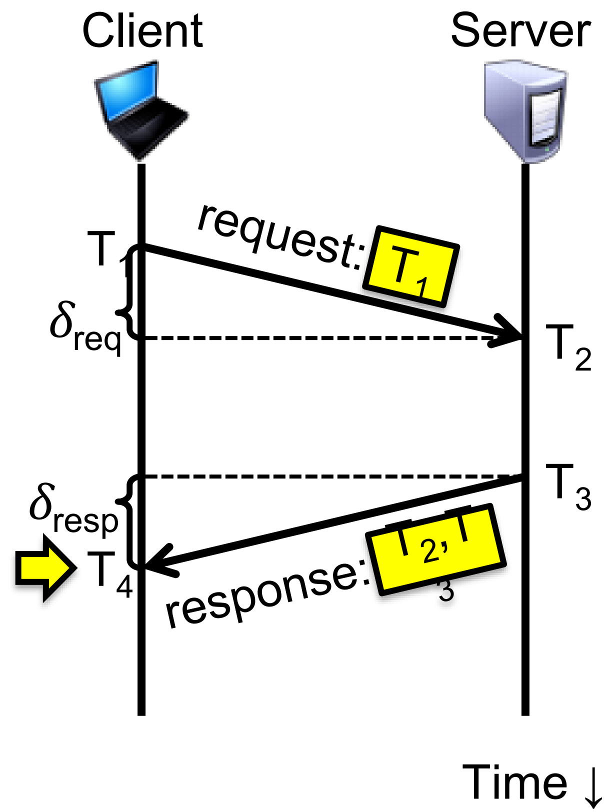
Client samples round trip time (δ)

$\delta = \delta_{req} + \delta_{resp} = (T_4 - T_1) - (T_3 - T_2)$

But client knows δ, not δ_{resp}

Assume: $\delta_{req} \approx \delta_{resp}$

Client sets clock ← $T_3 + ½\delta$



Clock synchronization: Take-away points

Clocks on different systems will always behave differently

- Disagreement between machines can result in undesirable behavior

NTP clock synchronization

- Rely on timestamps to estimate network delays
- 100s μs−ms accuracy
- Clocks never exactly synchronized

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Often inadequate for distributed systems

- Often need to reason about the order of events
- Might need precision on the order of ns

Idea: Logical clocks

Landmark 1978 paper by Leslie Lamport

Insight: only the events themselves matter

Idea: Disregard the precise clock time
Instead, capture just a "happens before" relationship between a pair of events

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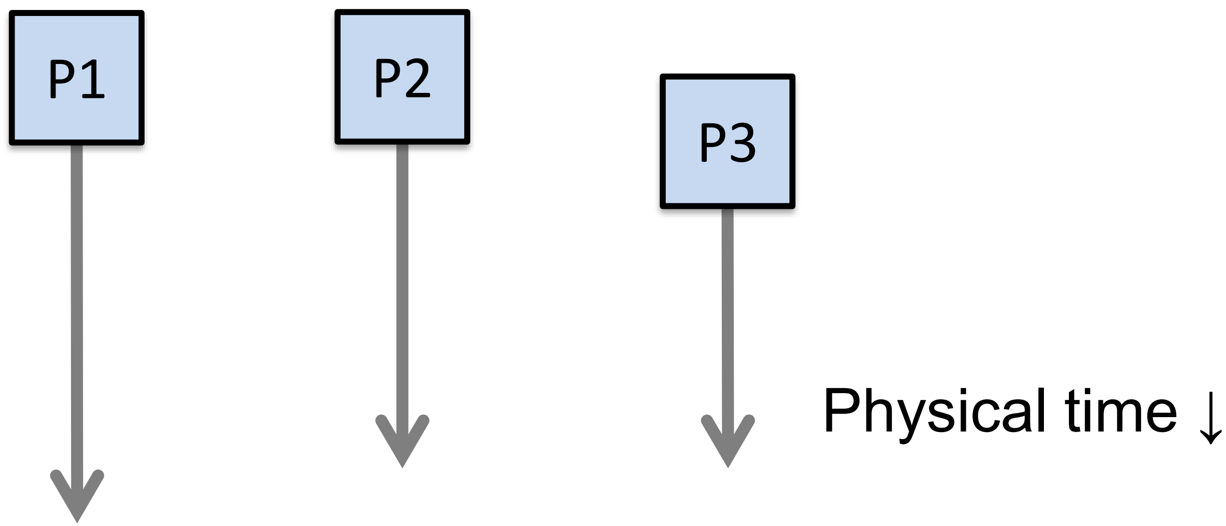
Defining “happens-before” (→)

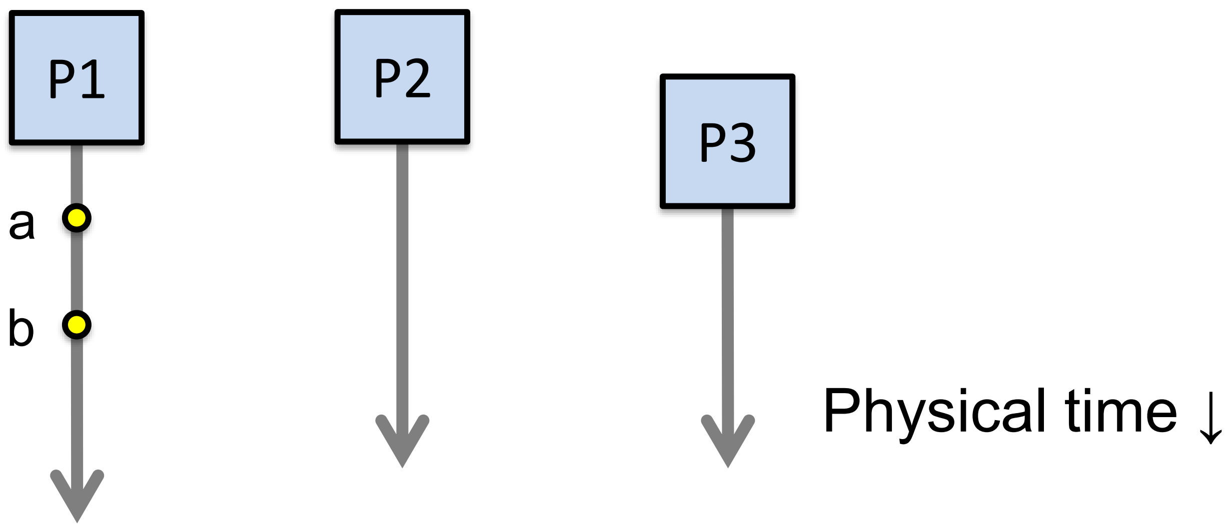
Consider three processes: P1, P2, and P3

Notation: Event a happens before event b (a → b)



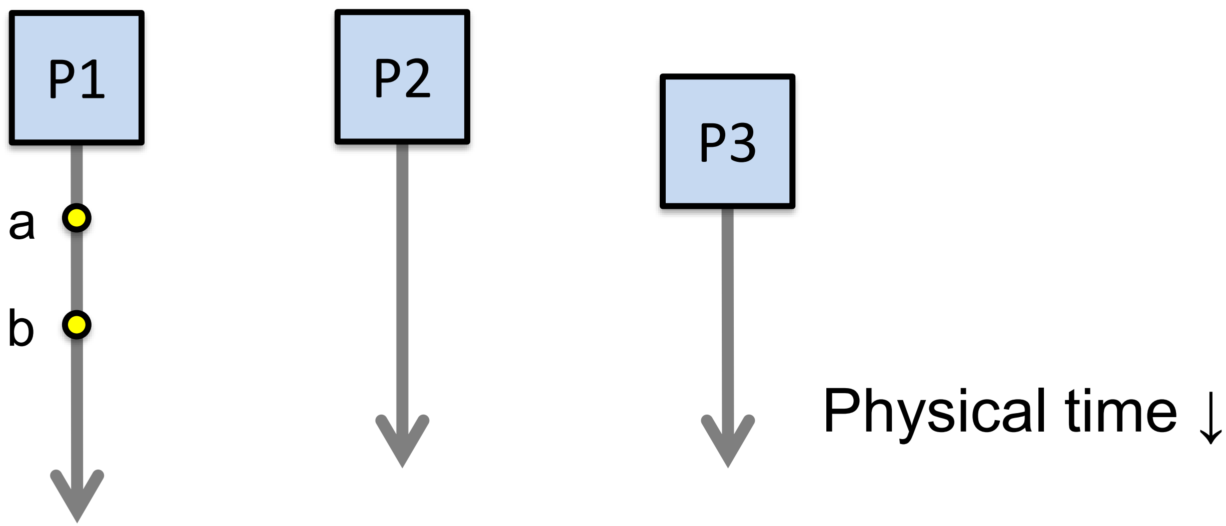
Defining "happens-before" (→)

Can observe event order at a single process



Defining “happens-before” (→)

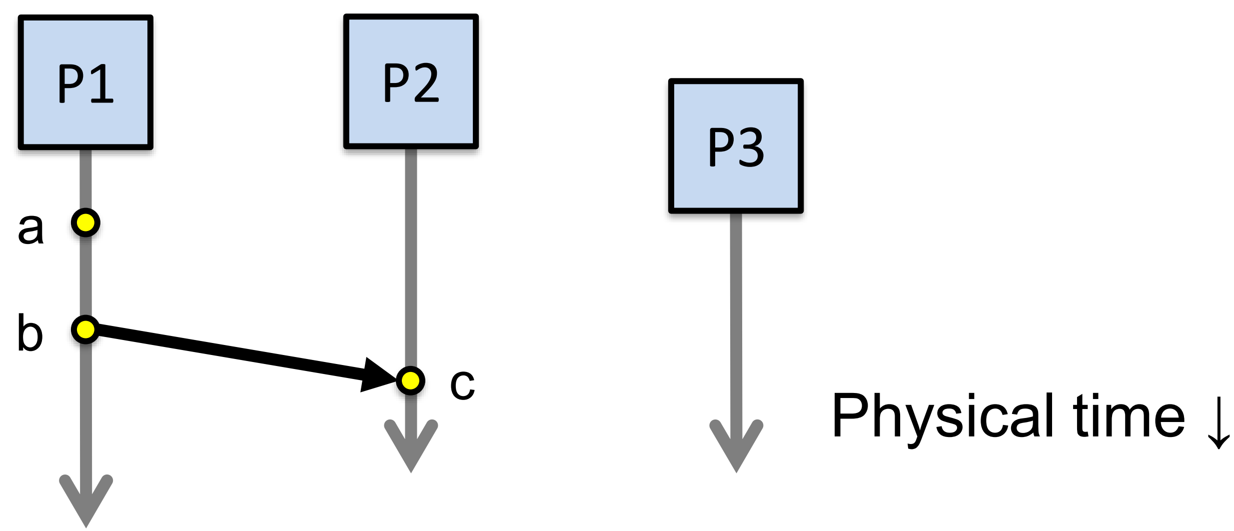
1. If **same process** and **a** occurs before **b**, then **a → b**



Defining "happens-before" (→)

1. If **same process** and **a** occurs before **b**, then **a → b**

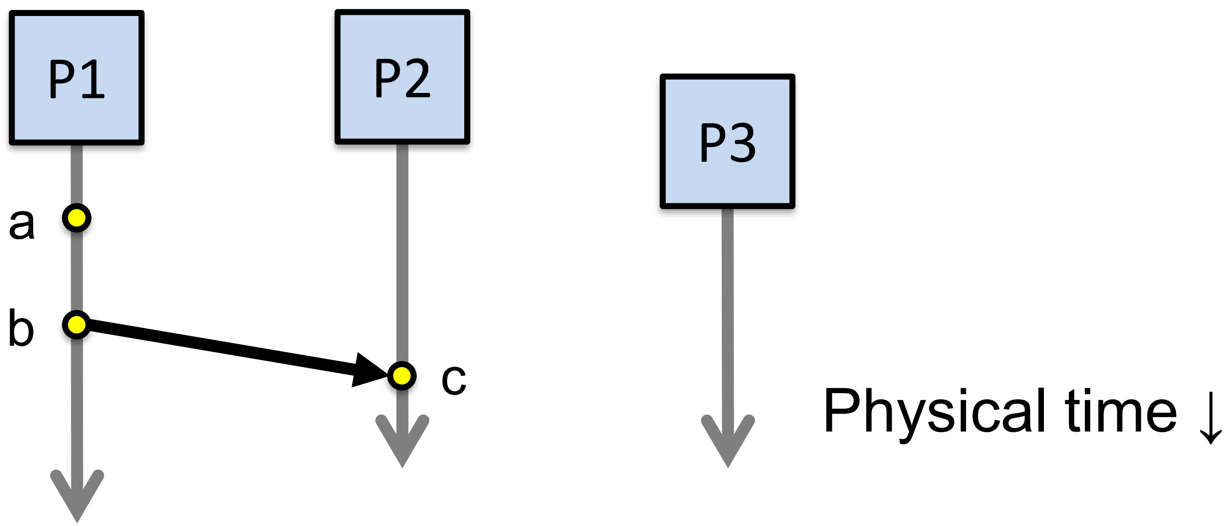
2. Can observe ordering when processes communicate



Defining "happens-before" (→)

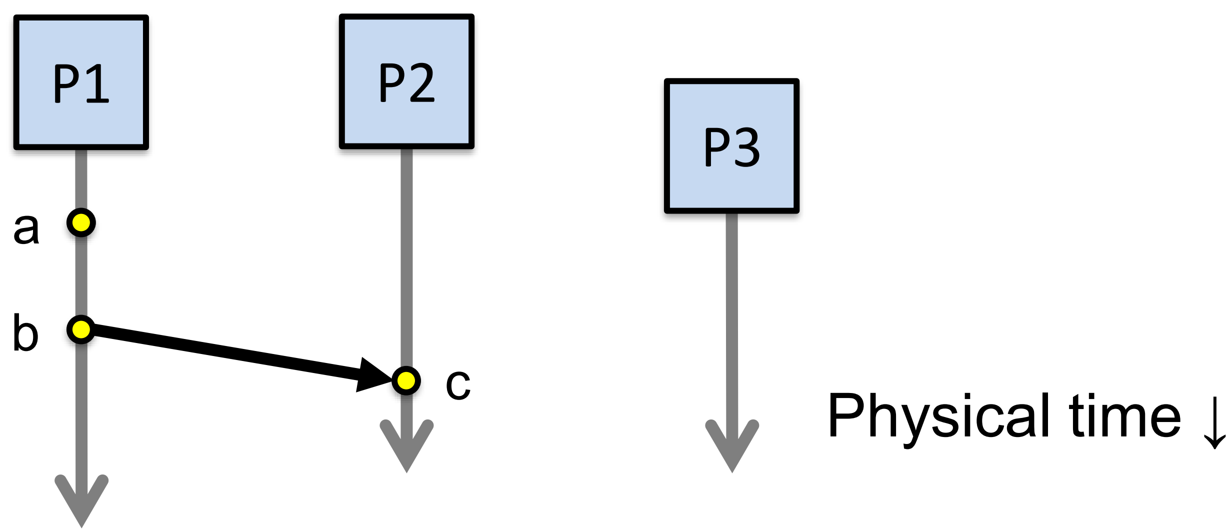
1. If **same process** and **a** occurs before **b**, then **a → b**

2. If **c** is a message receipt of **b**, then **b → c**



Defining “happens-before” (→)

1. If **same process** and **a** occurs before **b**, then **a → b**
2. If **c** is a message receipt of **b**, then **b → c**
3. Can observe ordering transitively

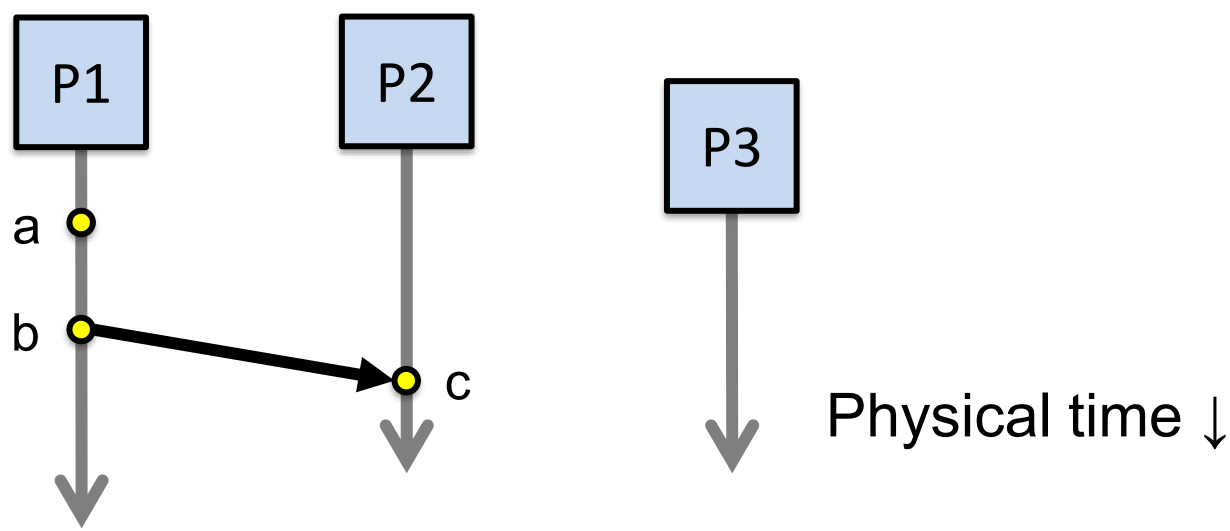


Defining “happens-before” (→)

1. If **same process** and **a** occurs before **b**, then **a → b**

2. If **c** is a message receipt of **b**, then **b → c**

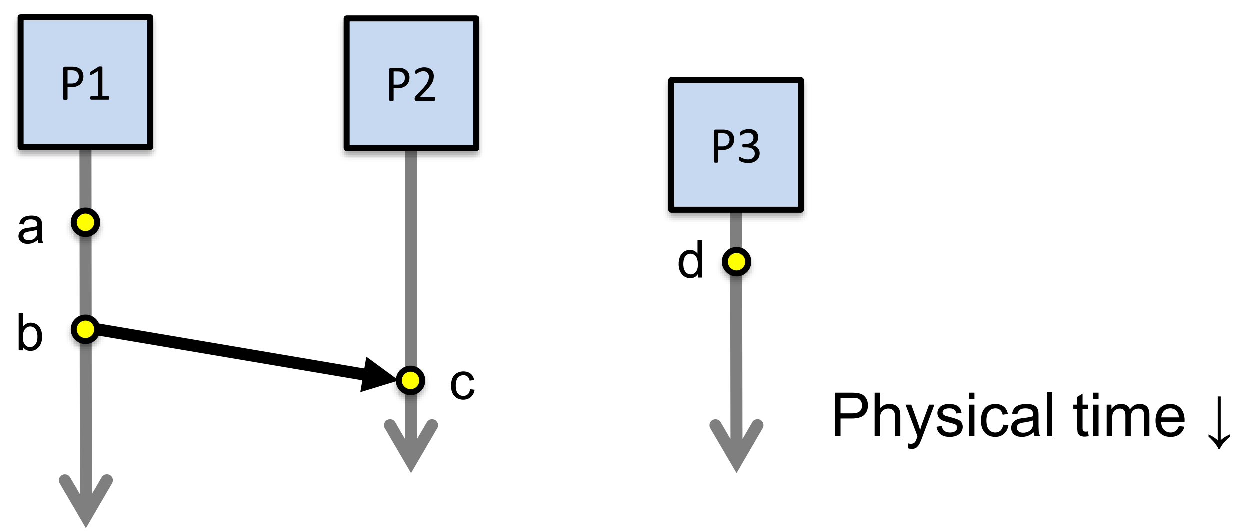
3. If **a → b** and **b → c**, then **a → c**



Concurrent events

Not all events are related by →

a, d not related by → so concurrent, written as a || d



Lamport clocks: Objective

We seek a clock time C(a) for every event a

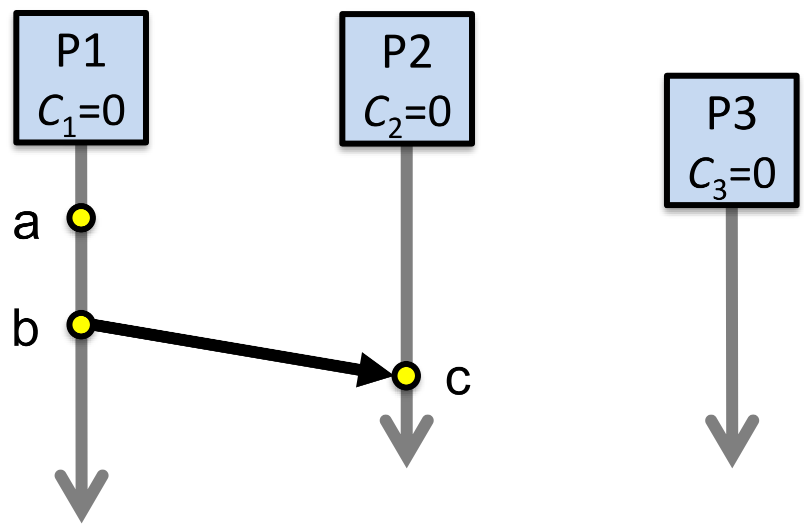
Plan: Tag events with clock times; use clock times to make distributed system correct

Clock condition: If a → b, then C(a) < C(b)

The Lamport Clock algorithm

Each process P_i maintains a local clock C_i

1. Before executing an event, $C_i \leftarrow C_i + 1$

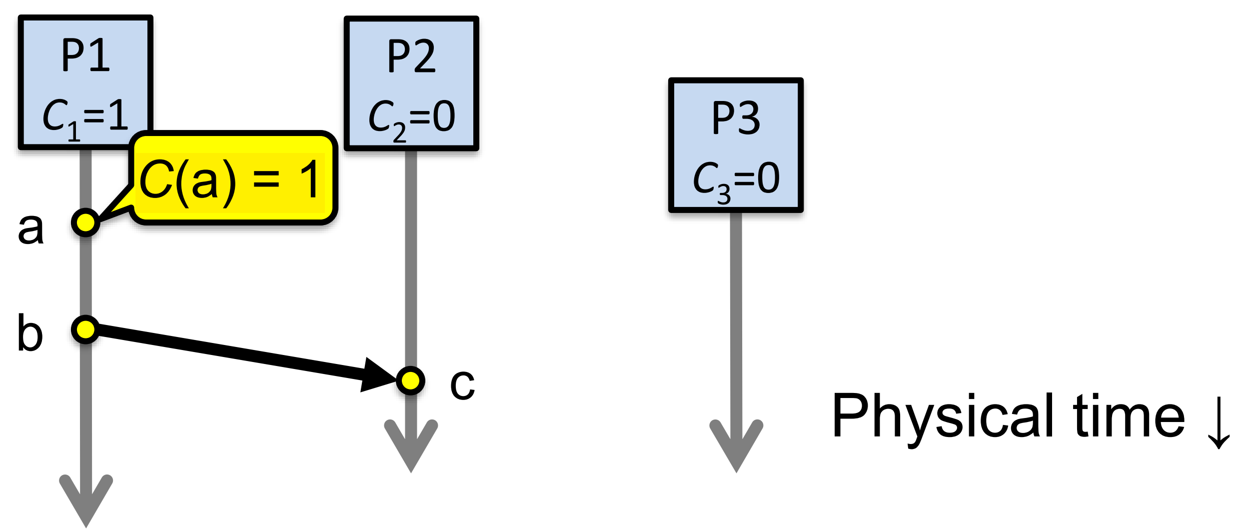


Physical time ↓

The Lamport Clock algorithm

1. Before executing an event a, $C_i \leftarrow C_i + 1$:

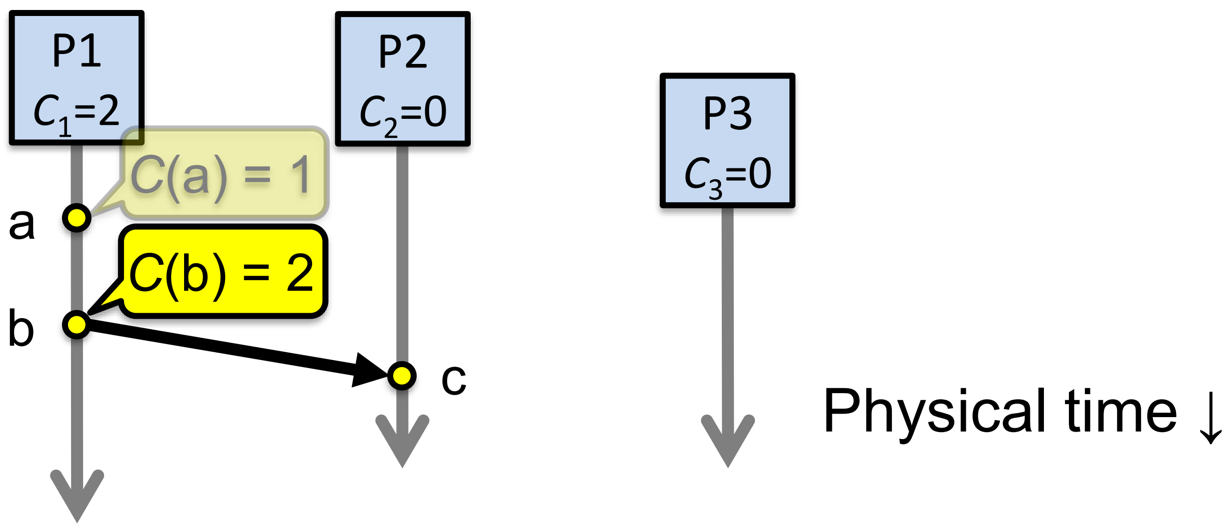
   Set event time $C(a) \leftarrow C_i$



The Lamport Clock algorithm

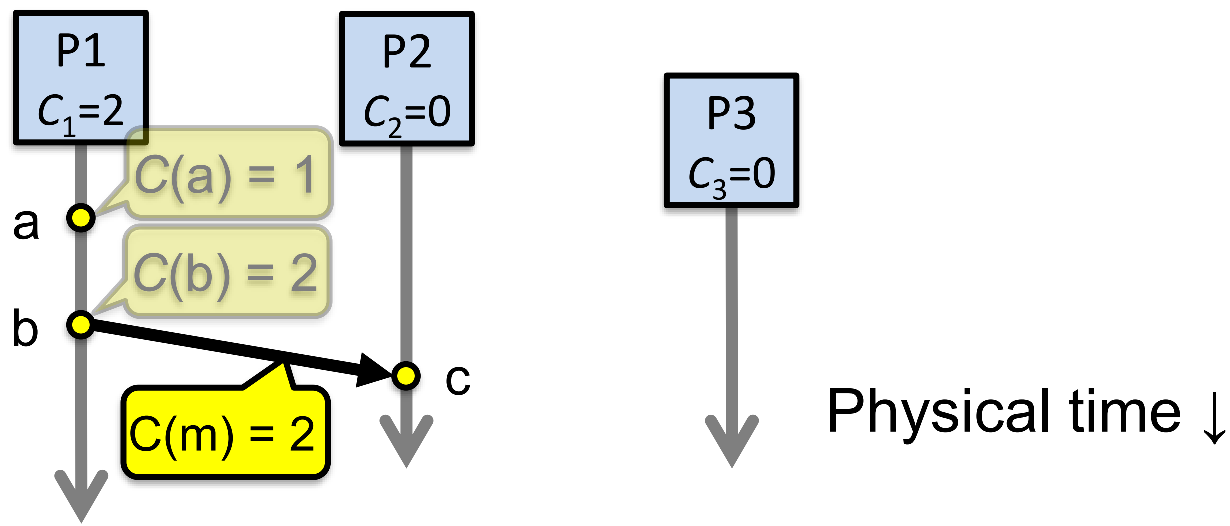
1. Before executing an event b, $C_i \leftarrow C_i + 1$:

   Set event time $C(\mathrm{b}) \leftarrow C_i$



The Lamport Clock algorithm

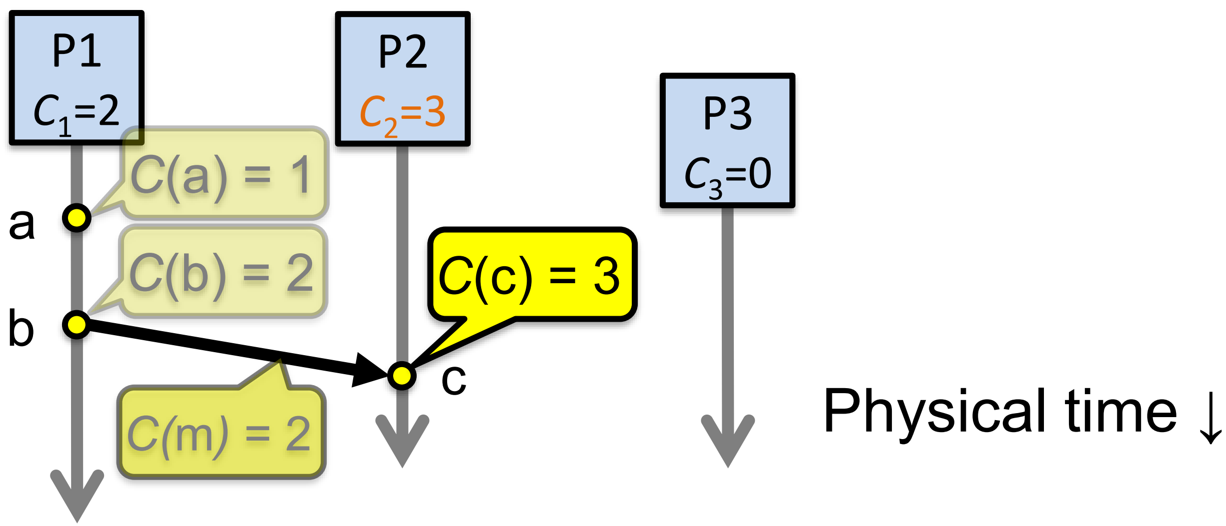
1. Before executing an event b, $C_i \leftarrow C_i + 1$
2. Send the local clock in the message m



The Lamport Clock algorithm

3. On process P_j receiving a message m:

Set C_j and receive event time $C(c) \leftarrow 1 + \max\{ C_j, C(m) \}$



Lamport Timestamps: Ordering all events

Break ties by appending the process number to each event:

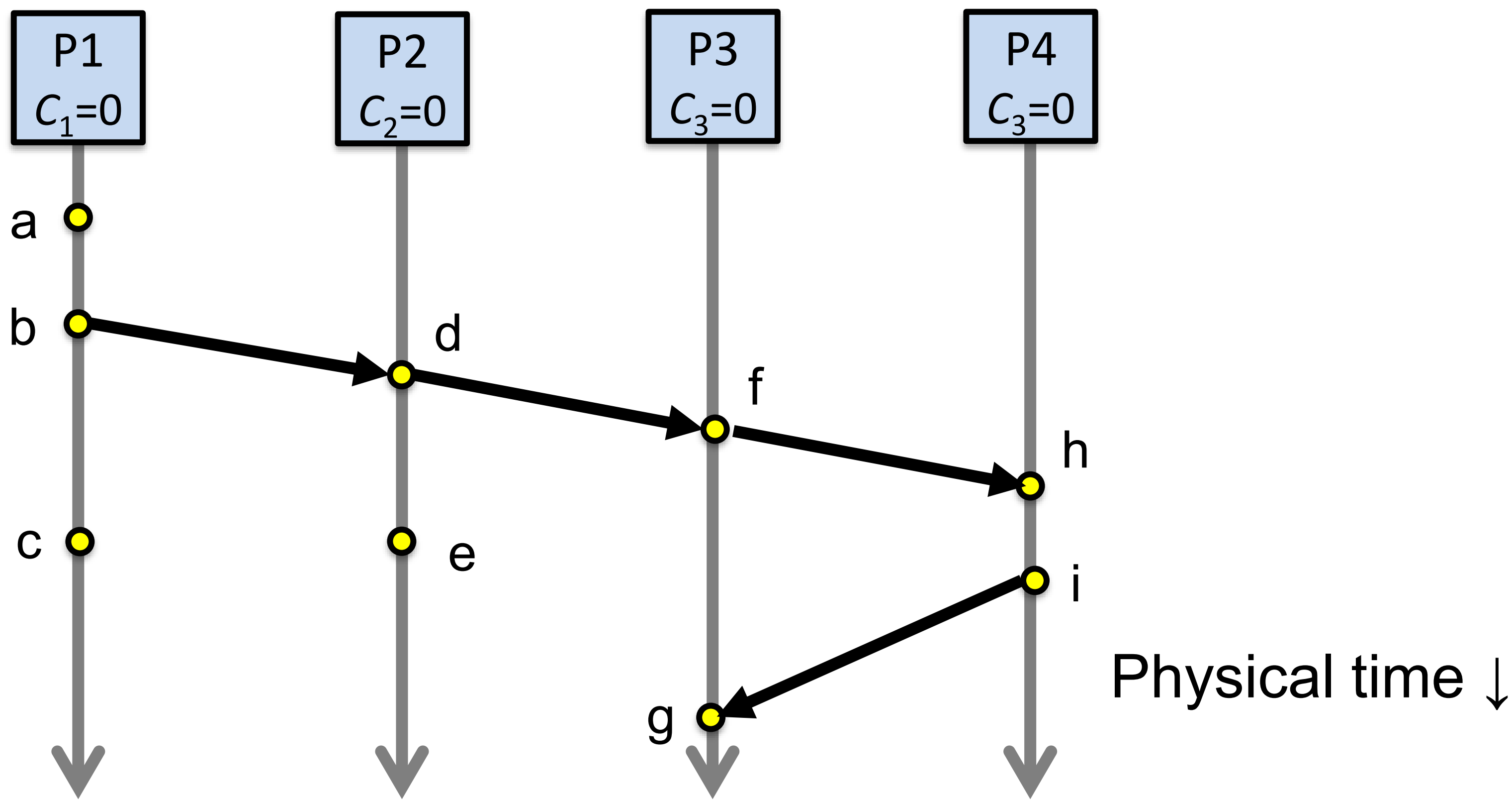
1. Process P_i timestamps event e with $C_i(e).i$

2. $C(a).i < C(b).j$ when:
   $C(a) < C(b)$, or $C(a) = C(b)$ and $i < j$

Now, for any two events a and b, C(a) < C(b) or C(b) < C(a)

This is called a total ordering of events

Order all these events



Totally-Ordered Multicast

Goal: All sites apply updates in (same) Lamport clock order

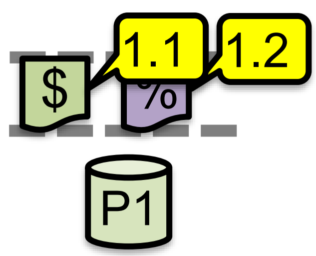
Client sends update to one replica site *j*

Replica assigns it Lamport timestamp $C_j \,.\, j$

Key idea: Place events into a sorted local queue

Sorted by increasing Lamport timestamps

Example: P1's local queue:



← Timestamps

Totally-Ordered Multicast (Almost correct)

1. On receiving an update from client, broadcast to others (including yourself)

2. On receiving an update from replica:
   a) Add it to your local queue
   b) Broadcast an acknowledgement message to every replica (including yourself)

3. On receiving an acknowledgement:
   Mark corresponding update acknowledged in your queue

4. Remove and process updates everyone has ack'ed from head of queue

Totally-Ordered Multicast (Almost correct)

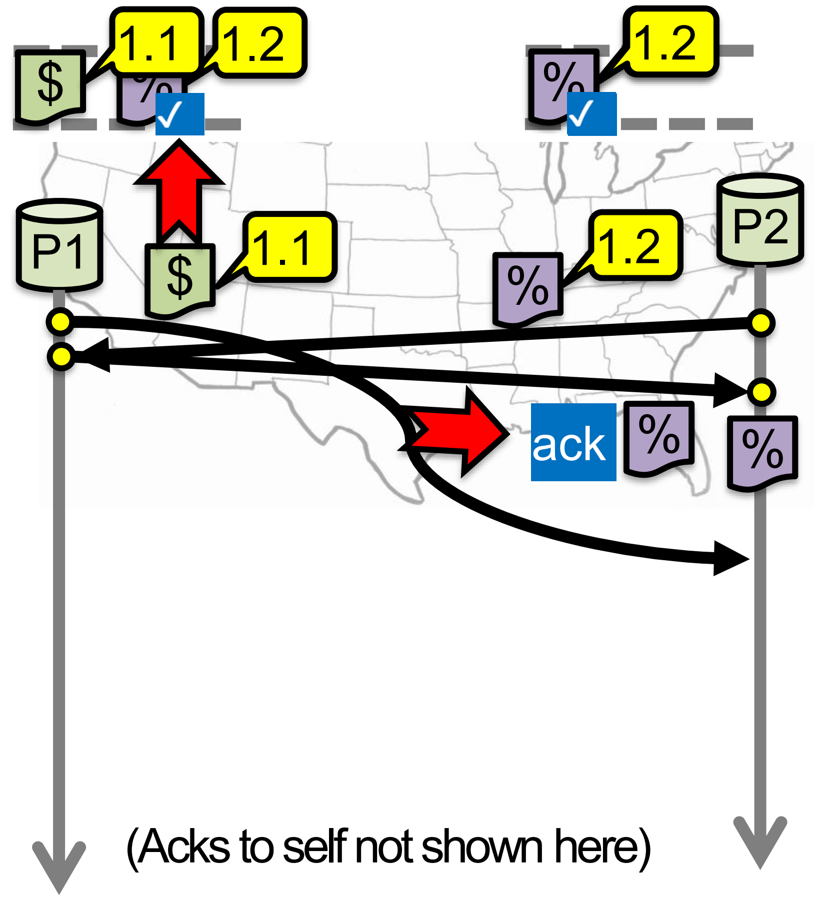
P1 queues $, P2 queues %

P1 queues and ack's %

P1 marks % fully ack'ed

P2 marks % fully ack'ed

✗ P2 processes %



Totally-Ordered Multicast (Correct version)

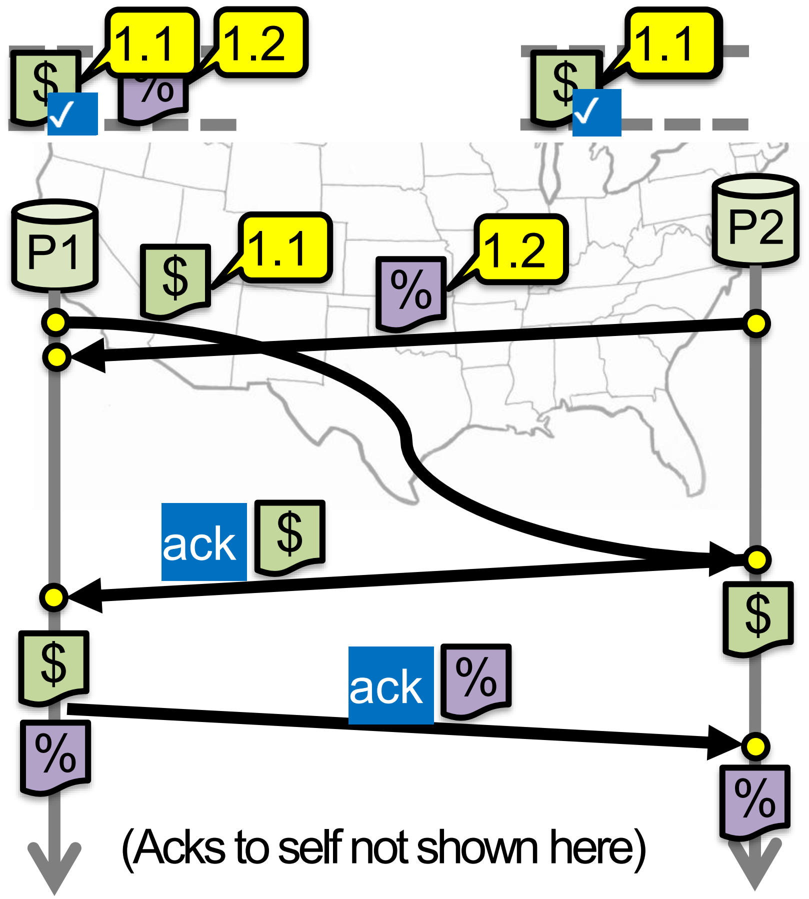
1. On receiving an update from client, broadcast to others (including yourself)

2. On receiving or processing an update:
   a) Add it to your local queue, if received update
   b) Broadcast an acknowledgement message to every replica (including yourself) only from head of queue

3. On receiving an acknowledgement:
   Mark corresponding update acknowledged in your queue

4. Remove and process updates <u>everyone</u> has ack'ed from <u>head</u> of queue

Totally-Ordered Multicast (Correct version)



So, are we done?

Does totally-ordered multicast solve the problem of multi-site replication in general?

Not by a long shot!

1. Our protocol assumed:
   - No node failures
   - No message loss
   - No message corruption
2. All to all communication does not scale
3. Waits forever for message delays (performance?)

Take-away points: Lamport clocks

Can totally-order events in a distributed system: that's useful!

We saw an application of Lamport clocks for totally-ordered multicast

But: while by construction, a → b implies $C(a) < C(b)$,

The converse is not necessarily true:

$C(a) < C(b)$ does not imply a → b (possibly, a || b)

Can't use Lamport timestamps to infer causal relationships between events

Vector clock: Introduction

One integer can't order events in more than one process

So, a Vector Clock (VC) is a vector of integers, one entry for each process in the entire distributed system

- Label event e with $VC(e) = [c_1, c_2 \ldots, c_n]$
  - Each entry c_k is a count of events in process k that causally precede e

Vector clock: Update rules

Initially, all vectors are [0, 0, …, 0]

Two update rules:

1. For each local event on process i, increment local entry c_i

2. If process j receives message with vector $[d_1, d_2, \ldots, d_n]$:
   Set each local entry $c_k = \max\{c_k, d_k\}$
   Increment local entry c_j

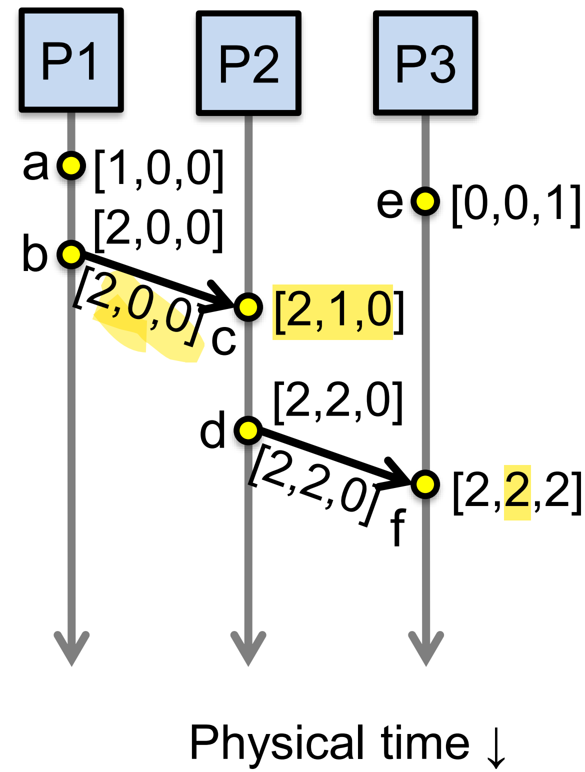
Vector clock: Example

All processes' VCs start at [0, 0, 0]

Applying local update rule

Applying message rule

- Local vector clock piggybacks on inter-process messages



Comparing vector timestamps

Rule for comparing vector timestamps:

$V(a) = V(b)$ when $a_k = b_k$ for all k

$V(a) < V(b)$ when $a_k \leq b_k$ for all k and $V(a) \neq V(b)$

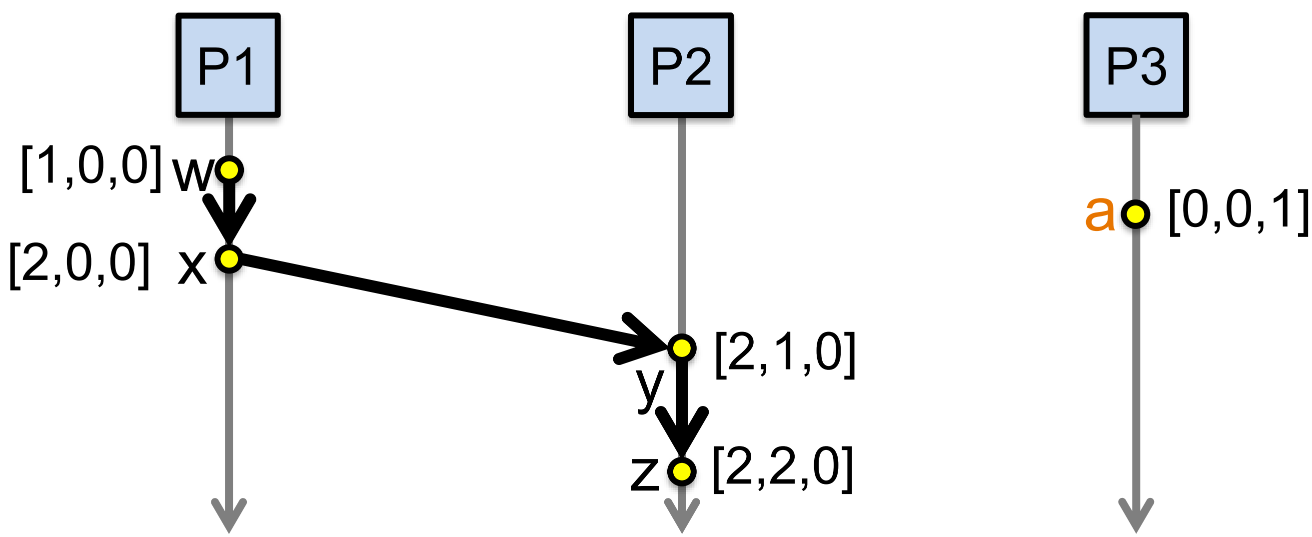
Concurrency:

$V(a) \parallel V(b)$ if $a_i < b_i$ and $a_j > b_j$, some i, j

Vector clocks capture causality

V(w) < V(z) then there is a chain of events linked by Happens-Before (→) between w and z

V(a) || V(w) then there is no such chain of events between a and w



Comparing vector timestamps

Rule for comparing vector timestamps:

- $V(a) = V(b)$ when $a_k = b_k$ for all k
  - They are the same event
- $V(a) < V(b)$ when $a_k \le b_k$ for all k and $V(a) \ne V(b)$
  - $a \rightarrow b$

Concurrency:

- $V(a) \parallel V(b)$ if $a_i < b_i$ and $a_j > b_j$, some i, j
  - $a \parallel b$

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Two events a, z

Lamport clocks: C(a) < C(z)

Conclusion: z -/-> a, i.e., either a → z or a || z

Vector clocks: V(a) < V(z)

Conclusion: a → z

Vector clock timestamps precisely capture happens-before relation (potential causality)

Resources

Slides: courtesy of **Prof. Dr. Boris Koldehofe** who developed the previous version of the slides, which were adapted from similar courses offered at the University of Stuttgart (**Prof. Dr.Kurt Rothermel**), and EPFL (**Prof. Dr. Rachid Guerraoui**).

Textbooks

- Ch. Cachin, L. Rodrigues, and R. Guerraoui: Introduction to Reliable and Secure Distributed Programming, 2nd Edition, 2011
  - Chapter 2